Solutions for Virtual Laboratory Tools for Hydrogen Training

2026 Trello Board

Product Backlog (Remaining Tasks):

- **As an admin, I want an in-tool editor, so I can edit the content of the training when needed.**
  Partially completed (Most major editors complete)
  Acceptance Criteria:
  - I can open the editor for an content element directly from the page that content is located.
  - I can freely edit a section of text.
  - After editing a section of text, I can commit the changes I made directly to the file storing the content.
  - I can freely change an image used in the content.
  - I can exit out of the editor without any changes I made being committed to the files storing the content.

- **As an admin, I want to be able to lock modules and quizzes behind progression through others, so I can prevent users from skipping content.**
  Partially completed (Quiz lock feature implemented)
  Acceptance Criteria:
  - Quiz unlocking tied to its modules
  - Sets of modules unlock sequentially
  - Enforced server-side, not just in UI
  - Locked state shows in UI
  - Admins are not required to complete prerequisite content

- **As a user, I want a Light/Dark Mode option so I can use the tool in a visual style I am comfortable viewing.**

- **As a user, I want training on safety protocols used in hydrogen laboratories, so I can safely navigate such environments.**
  Acceptance Criteria:
  - I learn the industry standard safety protocols
  - I learn what should be done in case of a safety hazard or risk identified in the lab environment

- **As a user, I want training on emergency procedures used in hydrogen laboratories, so I can properly act when present in such environments during emergencies.**

- **As a user, I want to be assessed on my knowledge of safety protocols in a hydrogen laboratory, so I can prove my understanding.**

- **As a user, I want to be assessed on my knowledge of emergency procedures in a hydrogen laboratory, so I can prove my understanding.**

- **As a user, I want to explore an example of an industrial environment that uses hydrogen fuel with explanations of possible hazards, so I can identify those hazards.**

- **As a user, I want to understand how to safely navigate and operate within a hydrogen lab environment without being physically present in one.**
  Acceptance Criteria:
  - The system provides an interactive lab environment for users to explore.
  - Users can navigate different areas within the lab.
  - Points of Interest are clearly visible or identifiable within the environment.
  - Users can select Points of Interest to view detailed explanations.
  - Each Point of Interest includes a clear description.

- **As a user, I want module page URLs to be name-based rather than number-based, so that they are easier to identify.**
  Utilises the 'slug' parameter of modules, rather than their id numbers. (e.g. /modules/hazards/gas-leak-detection, instead of /modules/hazards/1)

Done (Completed Tasks):

- **As a user, I want to explore an example of a laboratory that uses hydrogen fuel with explanations of possible hazards, so I can identify those hazards.**

- **As a user, I want to log in securely so that I can access my training dashboard and progress.**

- **As a user, I want training on risks and hazards associated with hydrogen equipment used in hydrogen laboratories, so I can safely operate such equipment.**

- **As a user, I want to receive a printable Proof-of-Completion, so I can show that I have completed the training.**

- **As an admin, I want to be able to track the progress of users of this tool so I can understand the knowledge level and preparedness of said students.**

- **As a user, I want an introductory training module on hydrogen, so I can understand the topic before directly handling the substance.**
  Became the Landing page.

- **As a user, I want to be assessed on my knowledge of hydrogen hazards in a hydrogen laboratory, so I can prove my understanding.**

- **Implement Unit & Integration Testing**

- **As a user, I want an About page briefly explaining the purpose of the tool, so that I can use it with full understanding.**

- **As a user, I want navigation links to pages not accessible unless logged in to be hidden when I am not logged in, so I can identify pages I can visit without having to log in.**

- **As a user, I want to navigate modules, scenarios and quizzes from a dashboard, so I can track my training and continue my learning.**

- **As a user, I want to be able to watch embedded videos in the Modules and Scenarios pages so I can learn more through an additional medium.**

- **As a user, I want quizzes to have a question pool so that they are more varied and challenging.**

- **As a user, I want a Review page so I can offer feedback on my training experience.**

- **As a user, I want an opt-in scoreboard page so I can choose to compare my quiz scores to those of my fellow users.**

- **As an admin, I want to be able to delete users from the administration page, as it would be easier and more convenient than doing so from the backend services that manage those user profiles.**

- **As an admin, I want to be able to print out users' progress/results in the form of an Excel spreadsheet, so I can integrate this data into Moodle.**

- **As a user, I want a hub page for both the modules and simulation pages, so I can more easily navigate them.**

- **As a user, I want the font size to be larger and the colours used to be more diverse, so I can have a more readable experience.**